Lexical Access in Bilingual Speech Production: A Replication of Language Switching Effects

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Theoretical Frameworks & Objectives

The Bilingual Selection Debate

The Shared Dilemma: Bilingual speakers constantly activate candidates in both separate lexicons simultaneously when presented with a single concept or image.

Hypothesis A (Language-Specific Selection): The Semantic System activates lexical nodes in both languages, but the subsequent selection mechanism is strictly language-specific. It only filters or "looks at" candidates within the target cued lexicon.

Hypothesis B (Language-Independent Selection): The selection mechanism is completely non-specific. Lexical nodes compete globally across both languages based on activation thresholds, requiring active inhibition of the non-target language.

Insight from Switch Costs: Measuring the temporal penalty of switching languages (Switch Cost) serves as a direct window into the cognitive work, inhibition, and executive control required to dynamically modulate these lexicon activation thresholds.

Background & Research Question

Costa & Santesteban (2004) found that L2 learners show *asymmetric* costs (harder to switch into dominant L1), while highly proficient bilinguals show *symmetric* costs — even with large proficiency differences between languages.

Research question: Does proficiency modulate asymmetric switching costs in a first-letter keypress production task with English–Spanish bilinguals?

Data Collection & Engineering

Task design: Language cue (English or Spanish) displayed for SOA duration (500ms or 800ms), followed by a pictured object. Participant pressed the first letter of the object name in the cued language. Switch condition coded as *repeat* (same language as previous trial) vs. *switch*.

Software stack: Task engineered and hosted using jsPsych (v7.3) for millisecond-accurate keyboard logging.

Technological pivot: Shifted from an acoustic voice-key setup (Java backend) due to micro-fluctuations in mic sensitivity and massive .wav file fragmentation. The keypress paradigm provided cleaner, more reliable RT data.

Data integrity filtering: Two exclusion criteria were applied. First, participants reporting an L1 other than English or Spanish (Italian, Portuguese, Polish, Slovene, Indonesian; n=11) were excluded from subsample analyses, as their dominant language did not match either target language. Second, neutral trials (the first trial per participant) were excluded as they cannot be classified as switch or repeat.

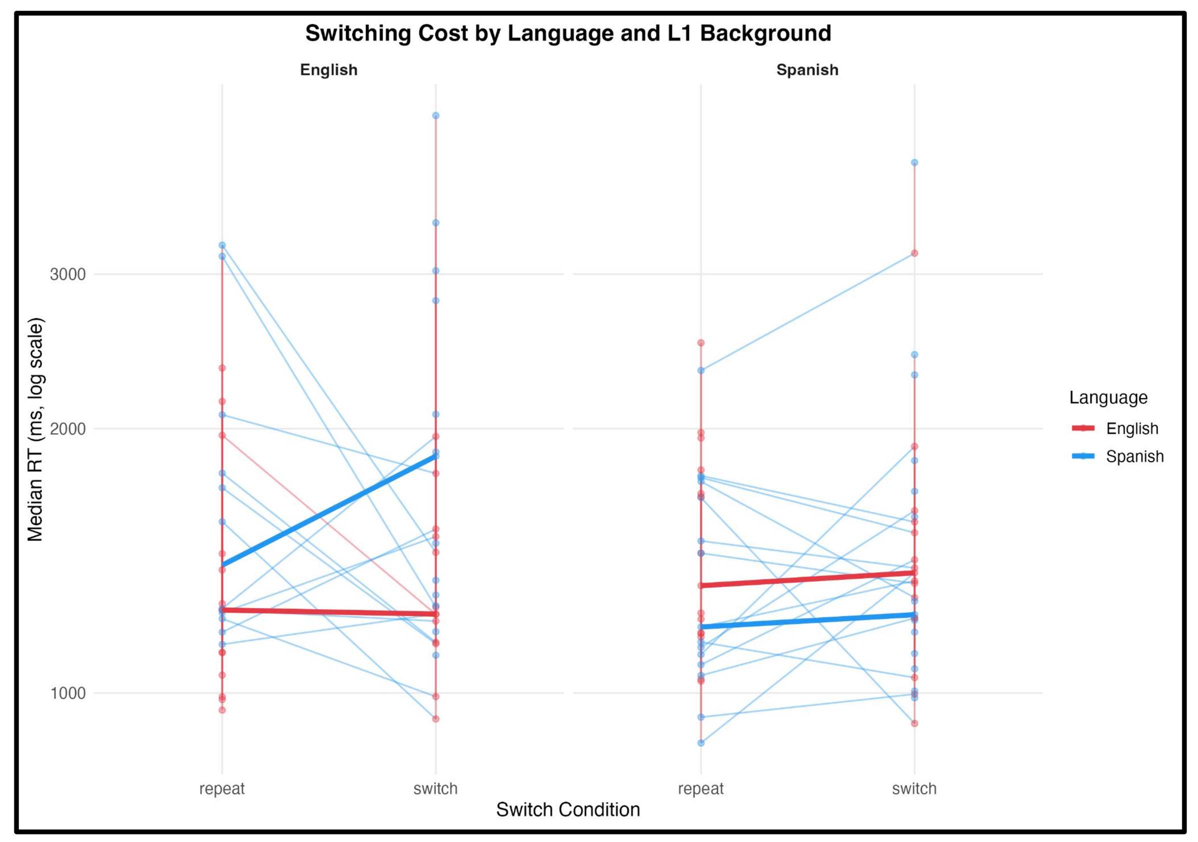
Final sample: 29 valid proficiency-mapped profiles (12 Native/Bilingual, 8 Advanced, 6 Intermediate, 2 Beginner) retained for subsample analyses; full sample of N=40 retained for task-level mechanics.

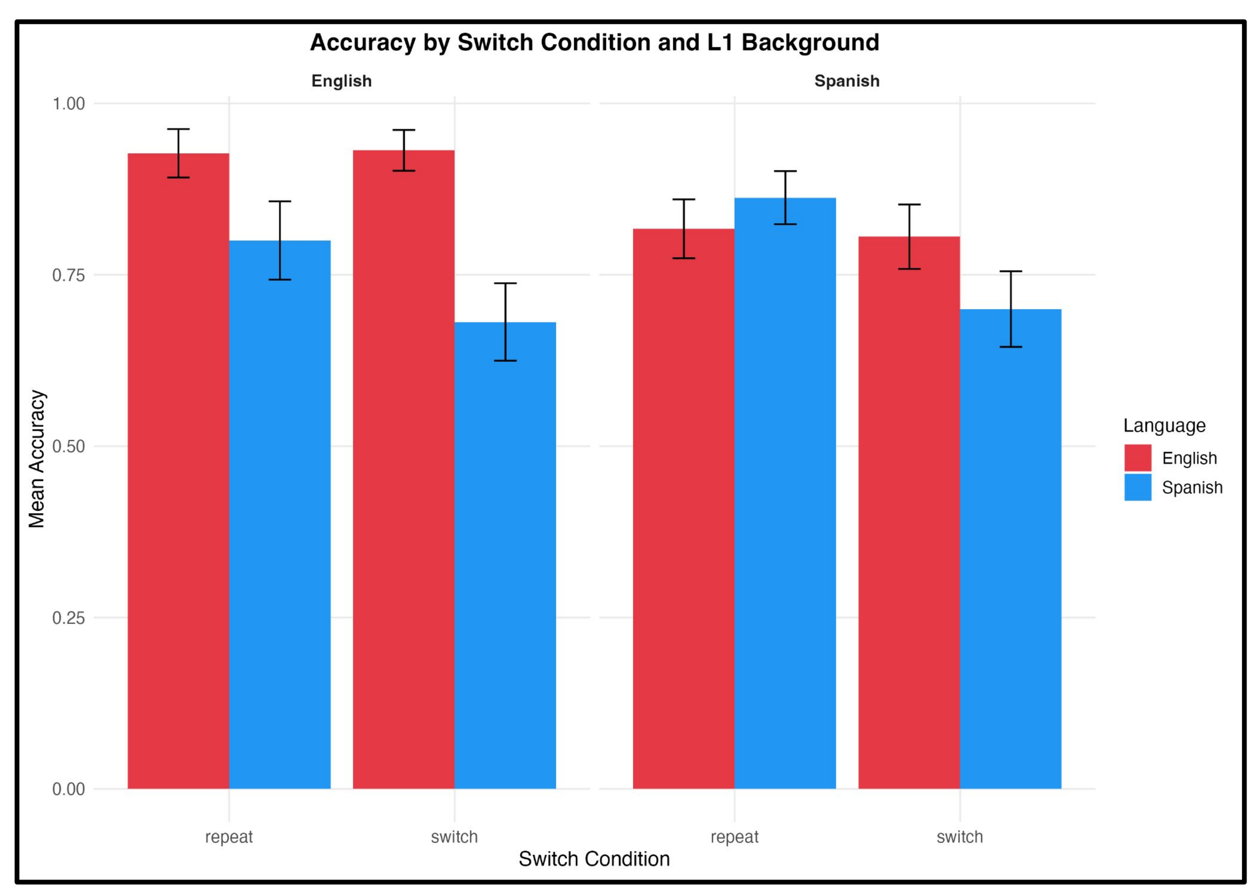
Analysis: Linear mixed effects models (lme4/lmerTest) on log-transformed RT; generalized linear mixed model (binomial) for accuracy. Participant included as random effect. Subsample analysis (n=29) restricted to English and Spanish L1 participants with L1 background.

Key Finding - L1 Background Effect

When the subsample (n=29) was divided by dominant language background, two distinct switching cost patterns emerged that directly map onto the profiles identified by Costa & Santesteban (2004). This is the central finding of the replication.

Direct replication of Costa & Santesteban (2004) Experiment 5: English L1 participants match the asymmetric cost profile expected of L2 learners; Spanish L1 participants match the symmetric cost profile of highly proficient bilinguals. This distinction emerges clearly in accuracy data and is consistent across both RT and accuracy analyses.

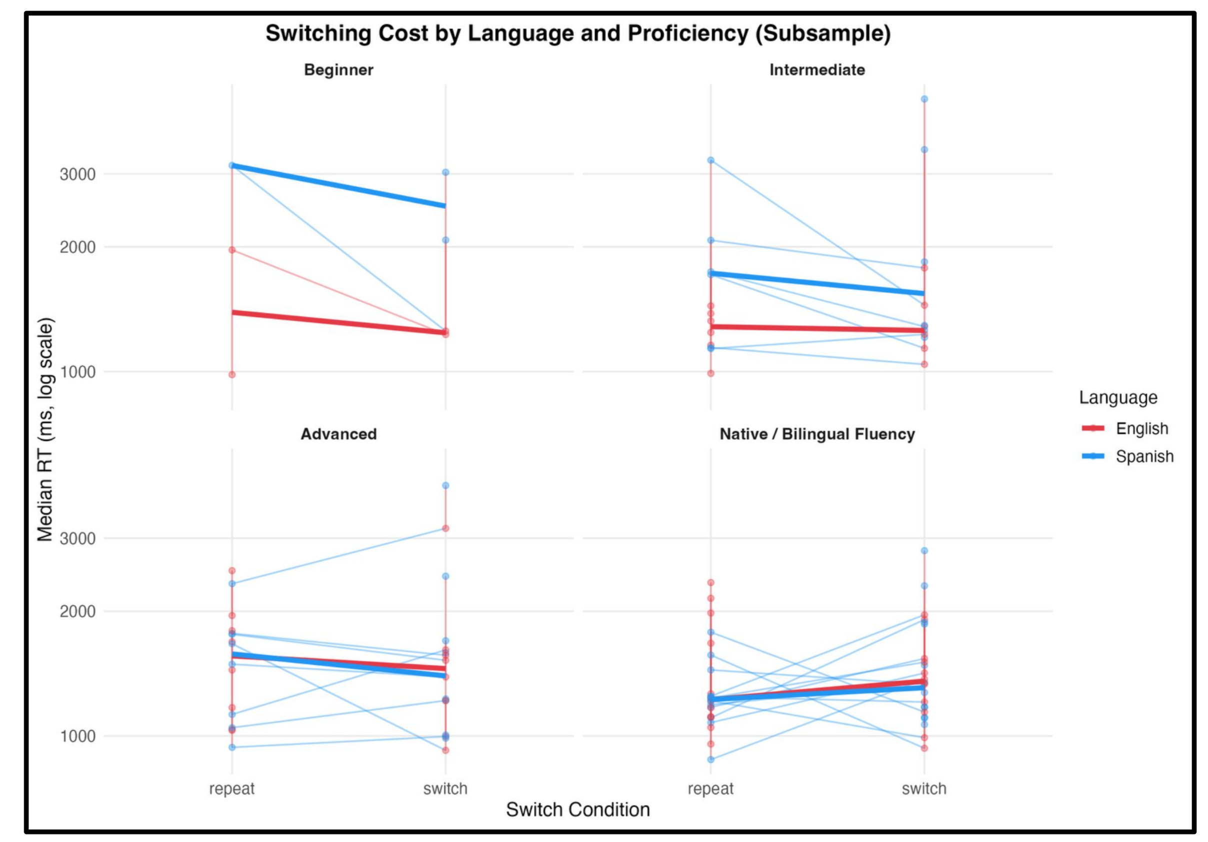


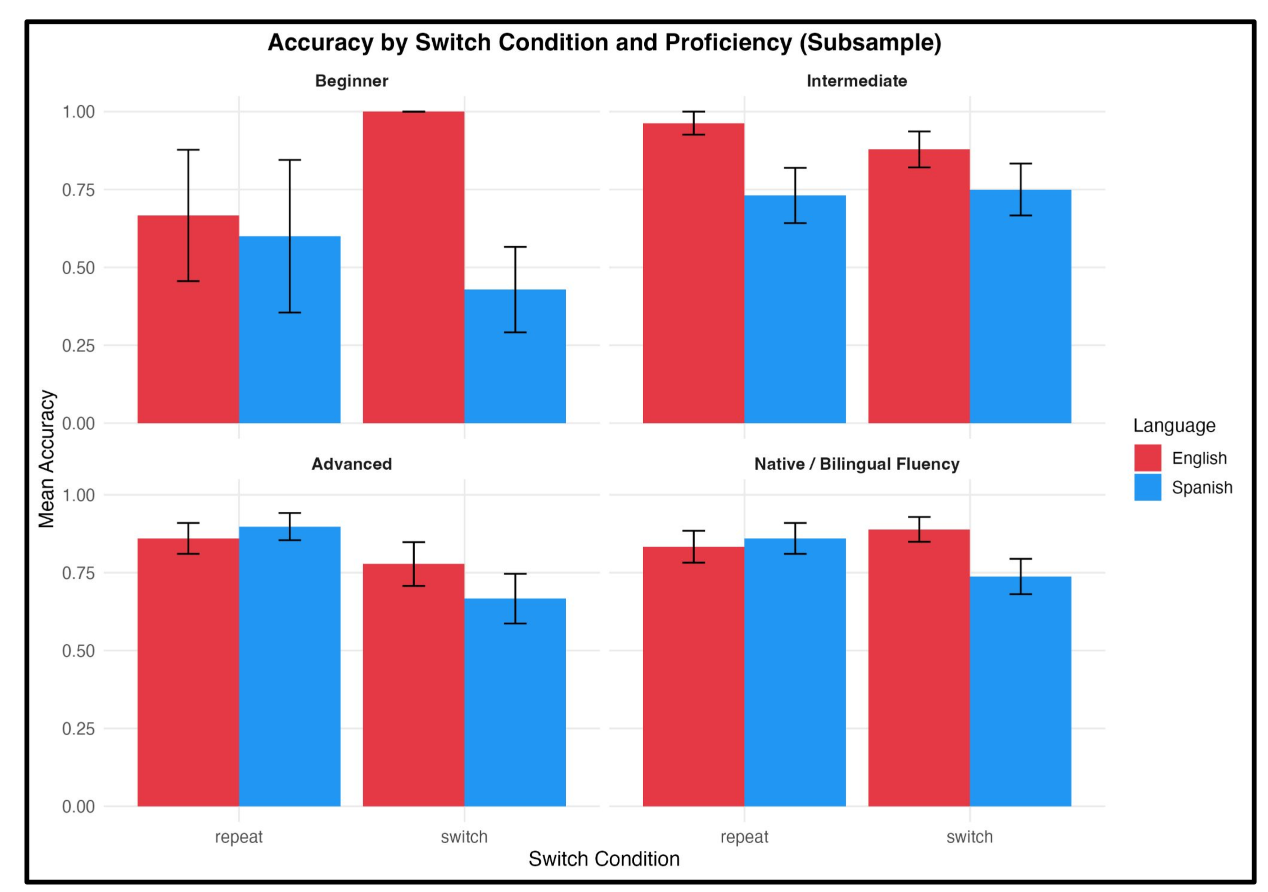


Key Finding - Proficiency Gradient

Subsample (n=29) split by self-reported proficiency reveals a monotonic gradient: switching cost asymmetry decreases as proficiency increases toward Native/Bilingual level, consistent with the inhibitory control framework.

Note: Proficiency main effects were not statistically significant in mixed effects models (likely due to small n per group), but the visual gradient is consistent with Costa & Santesteban's framework and the inhibitory control account.





Statistical Results

Linear mixed effects model (log RT) and generalized linear mixed model (binomial accuracy). Participant as random effect. Full sample N=40; subsample n=29.
Log RT used for linear models

Effect	RT	Accuracy	Sample
L2 Processing Cost	p < 0.001***	p < <0.001***	Both
Switch Cost	n.s.	p = 0.025*	Full only
Switch x Language	n.s.	p = 0.064	Full only
SOA Condition	n.s	n.s.	Both
Proficiency	n.s	n.s	Both
L1 Background	n.s	n.s	Subsample

Conclusion

L2 processing cost is robust: Participants were consistently slower and less accurate in L2 across all models and sample compositions. Strongest and most replicable finding in the dataset.

Switching cost found in accuracy: Significant overall switching cost in accuracy (p=.025, full sample). Not significant in RT — accuracy may be more sensitive to switching demands in this keypress paradigm.

Asymmetry replicates original study: English L1 → asymmetric cost; Spanish L1 → symmetric cost. Direct replication of Costa & Santesteban's L2 learner vs. proficient bilingual distinction.

Proficiency gradient observed: Asymmetry decreases monotonically from Beginner to Native/Bilingual — consistent with inhibitory control model, though not statistically significant due to small cell sizes.

SOA null — task modality matters: 500 vs. 800ms cue did not replicate original SOA effect, likely due to keypress vs. spoken response format. Future work could use larger sample and stricter participant screening.

Reference: Albert Costa, Mikel Santesteban,Lexical access in bilingual speech production: Evidence from language switching in highly proficient bilinguals and L2 learners,Journal of Memory and Language, Volume 50, Issue 4, 2004, Pages 491-511